

Construction Sensitivity of Shortest-Path Anisotropy Diagnostics in Geometric Graphs from Black-Hole Embeddings

Ariadna Uxue Palomino Ylla
Department of Physics, Nagoya University

Abstract

Geometric graphs provide flexible discretizations of sampled spaces, but graph observables can depend strongly on the sampling rule, edge construction, and choice of control. This makes it difficult to determine whether an apparent geometric signal is a property of the sampled space or of the graph-building protocol. We study this problem using graph discretizations of embedded spatial slices of static black-hole geometries as controlled benchmarks.

We focus on a shell-based shortest-path anisotropy statistic, $C_{\log}$, defined as the cubic mean deviation of logarithmic shortest-path multiplicities over finite graph-distance shells. Representative Schwarzschild/Flamm, Bardeen, and Hayward graphs show similar radial organization under the original calibrated construction. We then perform a paired ten-seed Schwarzschild/Flamm audit using five graph/control protocols: the original matched-flat KNN convention, strict radius-threshold controls, pure KNN graphs, mutual KNN graphs, and edge-count-matched radius graphs. In every run the black-hole graph and its controls reuse the same radial-angular sample, and correlations are computed from twelve common radial bins.

The audited result differs qualitatively from the earlier single-realization construction sweep. The mean absolute black-hole/control correlation gaps are 0.242 ± 0.161 , 0.688 ± 0.442 , 0.233 ± 0.199 , 0.422 ± 0.323 , and 0.681 ± 0.445 for the original, strict-radius, pure-KNN, mutual-KNN, and edge-matched-radius protocols, respectively (mean $\pm$ standard deviation over ten seeds). Radius-style controls therefore do not erase the separation; they give the largest average magnitudes in the audited sample. Signed gaps vary across realizations and change sign under some protocols.

Thus, the black-hole/control separation is neither graph-construction invariant nor realization independent. $C_{\log}$ is a reproducible finite-graph statistic when the sampling, edge rule, shell radius, binning, seed, and null model are specified, but the present evidence does not support interpreting it as a graph-independent curvature scalar. More broadly, the results motivate a matched-control audit framework in which geometric claims are attached to a complete diagnostic–construction–control protocol.

1 Introduction

Geometric graphs are widely used as finite representations of sampled metric spaces, spatial data, and physical or data-driven geometries [1, 2, 3]. However, a graph observable measured on such a representation is rarely a function of geometry alone. It can also depend on sampling density, boundary effects, edge rule, graph scale, and null model. Similar difficulties appear in thresholded and density-controlled network analyses, where changing the graph construction can change the interpretation of a measured network statistic [4, 5, 6]. For this reason, geometric interpretations of graph diagnostics require matched controls and explicit construction-sensitivity tests.

This issue is related, but not identical, to the problem of defining curvature on discrete structures. Classical discretizations such as Regge calculus encode curvature through a specific discretized geometric framework [7]. Graph and cell-complex notions such as Forman and Ollivier Ricci curvature provide alternative discrete analogues with different mathematical meanings

[8, 9]. The present work takes a complementary approach. We do not attempt to define a new discrete curvature tensor or scalar. Instead, we ask when a shortest-path graph diagnostic displays geometric organization and whether that organization survives stricter matched-control tests.

We use embedded spatial slices of black-hole geometries as a controlled setting for this question. These surfaces provide smooth radial benchmarks rather than new astrophysical predictions. The benchmark families are the Schwarzschild/Flamm embedding and two regular black-hole models, Bardeen and Hayward [10, 11, 12, 13]. Under the original fixed construction, representative graphs from all three families show closely aligned radial $C_{\log}$ profiles. This observation motivates, but does not by itself establish, a construction-independent geometric signal.

The diagnostic $C_{\log}$ measures the dispersion of logarithmic shortest-path multiplicities over a finite graph-distance shell. An earlier single-realization construction sweep suggested that the black-hole/control gap was large under the original matched-flat KNN control and almost vanished under radius-style controls. The purpose of the present revision is to test that interpretation with a paired multi-seed audit and a single canonical correlation pipeline. The audit shows that the earlier collapse was not representative: strict-radius and edge-matched-radius controls instead produce the largest mean absolute gaps, while realization-to-realization variability is substantial.

The scientific message is therefore not that one construction is uniquely correct. The separation depends jointly on the diagnostic, graph construction, and control. We distinguish three logically different questions: (i) reproducibility within a fixed protocol, (ii) sensitivity across protocols, and (iii) invariance under graph construction. The ten-seed experiment quantifies the first two and does not support the third.

The paper retains the Schwarzschild/Flamm, Bardeen, and Hayward family benchmarks from the preceding draft. The multi-protocol statistical audit is performed on Schwarzschild/Flamm, where complete paired raw data and quality controls are available. The Bardeen and Hayward figures are used as fixed-protocol geometric context; extending the same ten-seed construction matrix to those families is identified explicitly as the next validation step.

2 Graph benchmarks and matched controls

2.1 Embedding geometries

We consider geometric graphs sampled from embedded spatial slices of static black-hole benchmark geometries. For a static, spherically symmetric metric written as

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2d\Omega^2, \quad (1)$$

the equatorial spatial slice has line element

$$dl^2 = f(r)^{-1}dr^2 + r^2d\phi^2. \quad (2)$$

Embedding this slice as a surface of revolution in Euclidean three-space gives

$$dl^2 = \left(1 + \left(\frac{dz}{dr}\right)^2\right)dr^2 + r^2d\phi^2, \quad (3)$$

and therefore

$$\frac{dz}{dr} = \sqrt{\frac{1}{f(r)} - 1}. \quad (4)$$

The Schwarzschild case uses

$$f_{\text{Schw}}(r) = 1 - \frac{2M}{r}. \quad (5)$$

For the Bardeen regular black-hole family, we use

$$f_B(r) = 1 - \frac{2Mr^2}{(r^2 + g^2)^{3/2}}, \quad (6)$$

and for the Hayward family,

$$f_H(r) = 1 - \frac{2Mr^2}{r^3 + 2M\ell^2}. \quad (7)$$

The representative visual benchmarks use $M = 1/2$, $g = 0.2$, and $\ell = 0.2$.

2.2 Matched-flat controls

For each black-hole graph, matched-flat controls reuse the same radial and angular samples. The flat control removes the embedded height,

$$(r, \phi, z(r)) \longrightarrow (r, \phi, 0), \quad (8)$$

while preserving the radial-angular sampling. The control therefore asks how much of the diagnostic organization remains after replacing the curved embedding by a flat geometry with the same sampling information.

The graph-construction rule is part of the protocol. A diagnostic value is not interpreted independently of the edge rule and null model; every comparison is treated as a diagnostic–construction–control triple.

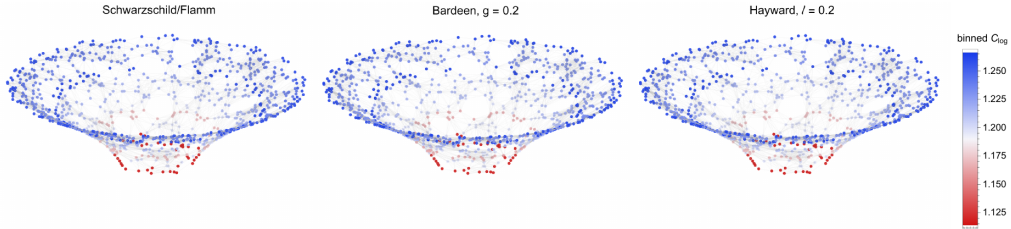


Figure 1: Representative Schwarzschild/Flamm, Bardeen ($g = 0.2$), and Hayward ($\ell = 0.2$) embedding graphs, colored by radially binned mean $C_{\log}$ on a common scale. These fixed-protocol visualizations motivate the broader family program; they are not a substitute for the paired multi-protocol audit reported below.

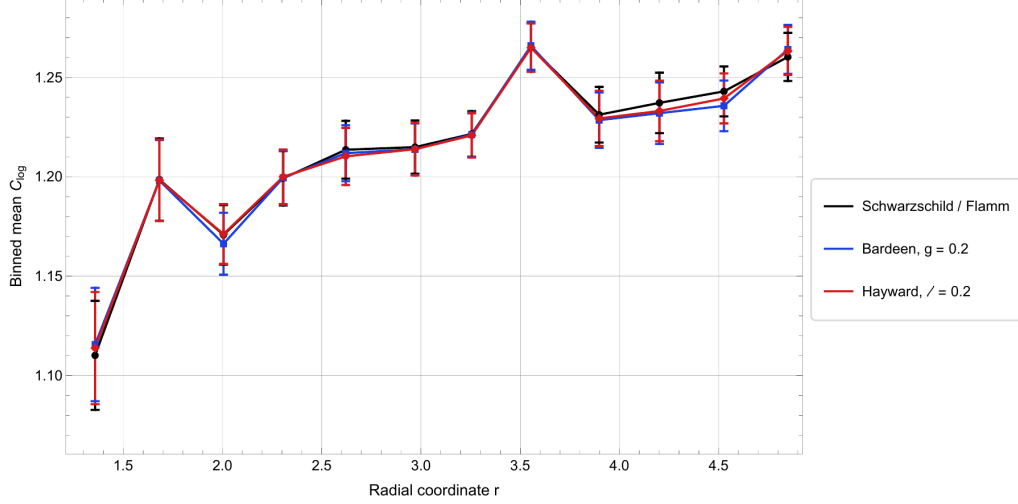


Figure 2: Representative binned radial $C_{\log}$ profiles for the three benchmark families under the original calibrated construction. The close alignment shows that the motivating radial organization is not visually unique to Schwarzschild, while the construction-invariance claim is tested quantitatively only in the audited Schwarzschild experiment.

3 Shortest-path anisotropy diagnostic

Let $G = (V, E)$ be a finite graph and let $d_G(i, j)$ denote graph distance. For a reference vertex i and graph-shell radius r_g , define

$$S_{r_g}(i) = \{j \in V : d_G(i, j) = r_g\}. \quad (9)$$

Let m_{ij} be the number of shortest paths between i and j . The statistic $C_{\log}$ is the cubic mean deviation of the logarithmic multiplicities over the shell,

$$C_{\log}(i; r_g) = \left(\frac{1}{|S_{r_g}(i)|} \sum_{j \in S_{r_g}(i)} \left| \log m_{ij} - \langle \log m \rangle_{S_{r_g}(i)} \right|^3 \right)^{1/3}. \quad (10)$$

This quantity probes anisotropy in shortest-path multiplicity patterns rather than degree or edge density alone. It is a finite-shell graph observable, not a continuum scalar by definition.

3.1 Canonical numerical protocol

The audited graphs use $M = 1/2$, $N = 1000$, construction parameter $k = 16$, graph-shell radius $r_g = 3$, twelve common radial bins, and seeds 1–10. One Schwarzschild/Flamm radial-angular sample is generated per seed and reused by all five graph/control protocols. The calibrated radius convention retains the factor 1.15 used in the original implementation.

Vertex-level $C_{\log}$ values are evaluated first. The black-hole and flat values are then binned on the same overlapping radial domain. Pearson correlations are computed between the common-bin mean radius and common-bin mean $C_{\log}$ profile,

$$\rho_{\text{BH}} = \text{Corr}(r, C_{\log})_{\text{BH}}, \quad \rho_{\text{flat}} = \text{Corr}(r, C_{\log})_{\text{flat}}, \quad (11)$$

with

$$\Delta\rho_r = \rho_{\text{BH}} - \rho_{\text{flat}}. \quad (12)$$

We report signed $\Delta\rho_r$ to retain direction and $|\Delta\rho_r|$ to summarize separation magnitude. No sign reversal to $-C_{\log}$ is used.

4 Paired construction-sensitivity audit

The audit tests whether the Schwarzschild/Flamm black-hole/control separation is stable under reasonable changes in the edge rule and flat-control construction. The five protocols are summarized in Table 1.

Table 1: Graph-construction and matched-control protocols used in the paired ten-seed audit.

Protocol	Black-hole graph	Flat control
Original matched-flat KNN	Calibrated radius-threshold graph	Matched-flat KNN graph
Strict radius	Calibrated radius-threshold graph	Radius-threshold graph from the flat sample
Pure KNN	Symmetrized KNN graph	Symmetrized KNN graph
Mutual KNN	Mutual-KNN graph	Mutual-KNN graph
Edge-matched radius	Calibrated radius-threshold graph	Radius graph matched to the black-hole edge count

All 50 runs passed the preregistered quality checks: ten rows per protocol, five rows per seed, twelve common bins, no missing correlations, maximum coordinate mismatch 1.33×10^{-15} , connected black-hole and flat graphs, unit largest-component and valid- $C_{\log}$ fractions, valid mutual-KNN subset checks, and exact or tie-explained edge matching. No seed was removed.

4.1 Protocol-level results

Table 2 reports the signed and absolute gaps. The result is not a monotone correction toward zero. Strict-radius and edge-matched-radius controls give the largest mean absolute separations, while the original and pure-KNN protocols give smaller averages. Strict radius and edge-matched radius are almost identical in paired mean absolute gap even though their control radii are selected differently.

Table 2: Ten-seed Schwarzschild/Flamm construction audit. Values are mean $\pm$ sample standard deviation. The 95% interval is a two-sided Student- t interval for the mean of $|\Delta\rho_r|$ with nine degrees of freedom.

Protocol	Signed $\Delta\rho_r$	$ \Delta\rho_r $	95% CI for mean $ \Delta\rho_r $
Original matched-flat KNN	0.160 ± 0.250	0.242 ± 0.161	[0.126, 0.357]
Strict radius	0.688 ± 0.442	0.688 ± 0.442	[0.372, 1.005]
Pure KNN	-0.189 ± 0.245	0.233 ± 0.199	[0.091, 0.375]
Mutual KNN	0.392 ± 0.362	0.422 ± 0.323	[0.192, 0.653]
Edge-matched radius	0.681 ± 0.445	0.681 ± 0.445	[0.363, 1.000]

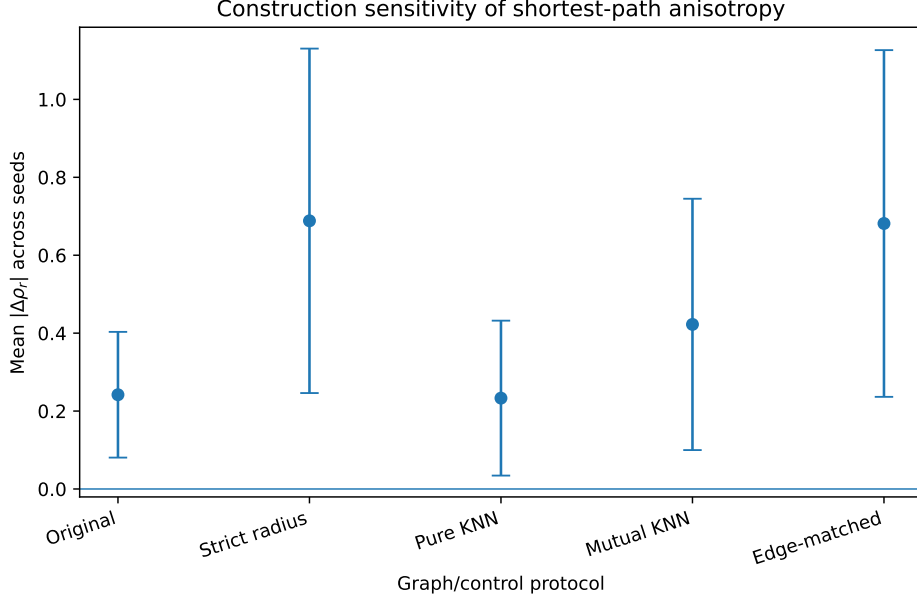


Figure 3: Mean absolute radial-correlation gap across the ten paired seeds. Error bars show sample standard deviation. The broad error bars are part of the result: protocol sensitivity and realization variability are both substantial.

The separation is not explained by a single edge-density comparison. Table 3 shows that strict radius and edge-matched radius give nearly the same gap even though the former has a small seed-dependent edge-count mismatch and the latter is exactly matched. Conversely, original and pure KNN have similar mean absolute gaps despite very different black-hole edge rules.

Table 3: Mean edge counts across ten seeds. The original protocol has a large black-hole/flat density mismatch, whereas the other symmetric protocols are approximately or exactly edge matched.

Protocol	Mean BH edges	Mean flat edges	Mean BH–flat difference
Original matched-flat KNN	10171.2	9044.4	1126.8
Strict radius	10171.2	10156.4	14.8
Pure KNN	9046.5	9044.4	2.1
Mutual KNN	6953.5	6955.6	-2.1
Edge-matched radius	10171.2	10171.2	0.0

4.2 Realization dependence and paired contrasts

Figures 4 and 5 display the same ten point samples across all protocols. The ordering is not identical from seed to seed. Original, pure-KNN, and mutual-KNN gaps change sign in some realizations; strict-radius and edge-matched-radius gaps are positive in all ten runs. The original protocol contains one conventional $1.5 \times \text{IQR}$ outlier at seed 10, $\Delta\rho_r = 0.6102$. Its graph, pairing, bins, connectivity, and estimator-validity checks all pass, so it is retained.

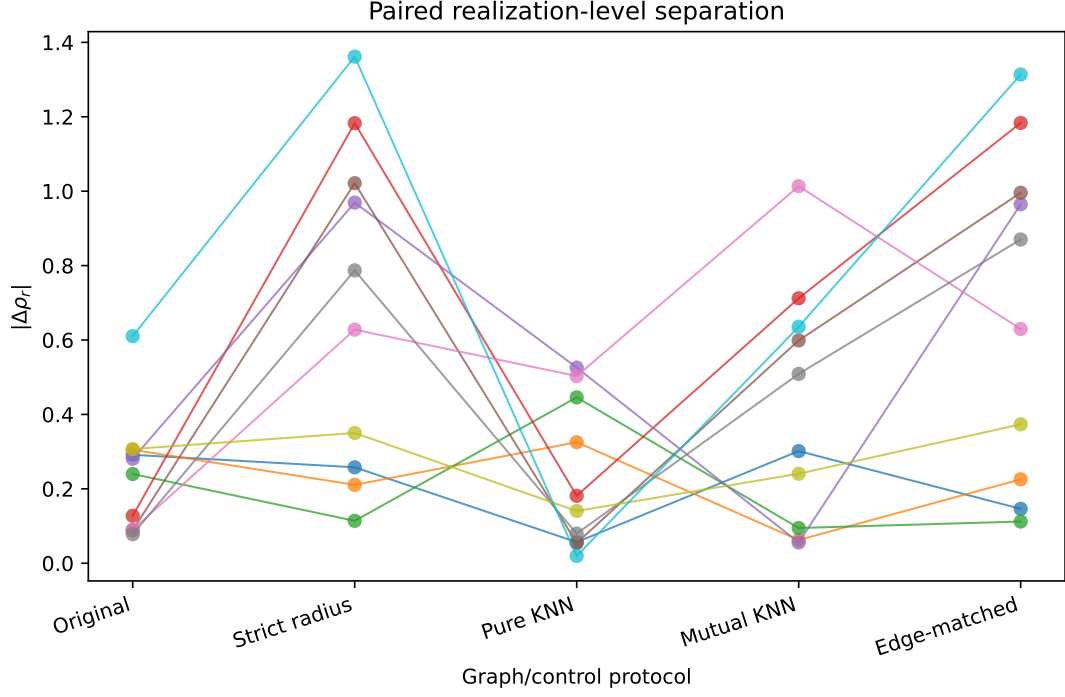


Figure 4: Paired $|\Delta\rho_r|$ values for all ten seeds. Lines join the same radial-angular sample across graph/control protocols.

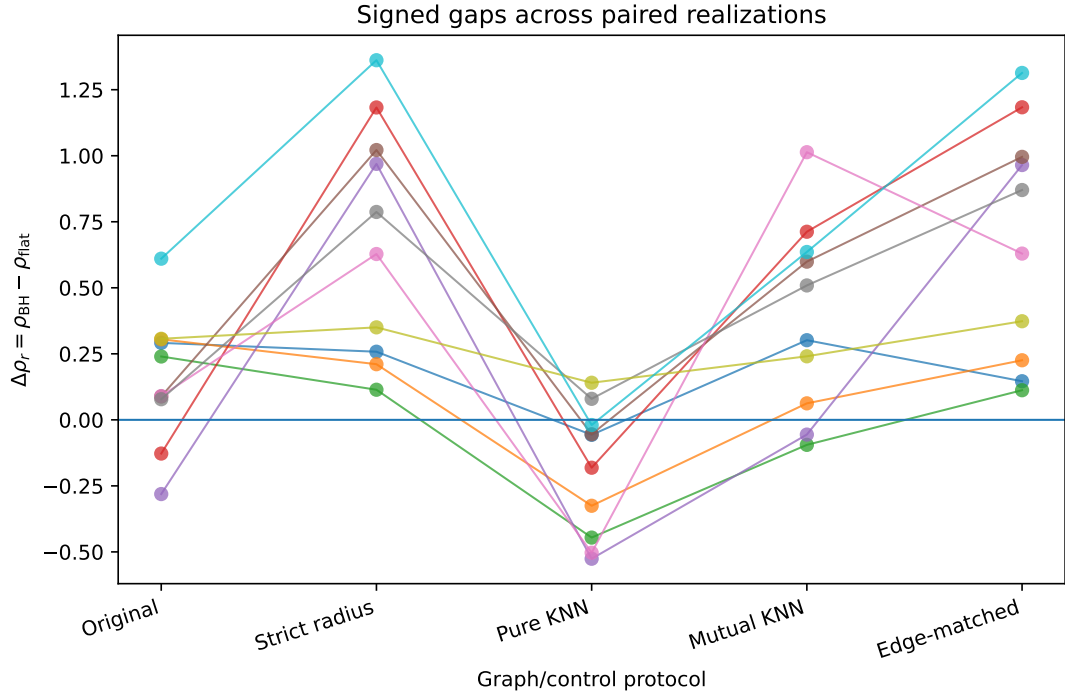


Figure 5: Signed $\Delta\rho_r$ for the same paired realizations. The sign changes show why signed and absolute gaps must be reported separately.

Selected paired contrasts are given in Table 4. The strict-radius and edge-matched-radius means exceed the original and pure-KNN means in this sample, while strict radius and edge-matched radius are mutually indistinguishable. These intervals quantify the paired experiment; they do not establish a universal ranking of graph constructions.

Table 4: Selected paired contrasts in $|\Delta\rho_r|$. The contrast is protocol A minus protocol B; intervals are two-sided Student- t intervals over ten paired seeds.

Protocol A	Protocol B	Mean paired difference	95% CI
Strict radius	Original matched-flat KNN	0.446	[0.122, 0.771]
Strict radius	Pure KNN	0.455	[0.071, 0.839]
Edge-matched radius	Original matched-flat KNN	0.440	[0.108, 0.771]
Edge-matched radius	Pure KNN	0.448	[0.066, 0.831]
Strict radius	Edge-matched radius	0.007	[−0.029, 0.043]
Original matched-flat KNN	Pure KNN	0.009	[−0.192, 0.209]

5 Relation to the broader black-hole benchmark program

Figures 1 and 2 remain useful because they show that the motivating radial organization appears in representative Bardeen and Hayward embeddings under the original fixed construction. The new audit changes the interpretation of that observation. Similar-looking profiles across geometries do not imply that the black-hole/control gap is invariant under changes in the edge rule or null model.

The earlier construction heatmap and table, which reported a common gap near 0.78 for all three families and a collapse to the 10^{-3} level under radius-style controls, were based on an incompatible single-realization/legacy pipeline. They are superseded by the canonical ten-seed Schwarzschild results in Table 2 and are not reproduced here. A direct cross-family construction claim requires the same paired common-bin audit for Bardeen and Hayward. This is a bounded extension of the present work rather than a change of research question.

An exploratory comparison with shell size, mean logarithmic multiplicity, degree, and clustering was included in the previous draft. Because those values were not regenerated under the canonical ten-seed pipeline, they are not used as inferential evidence here. The conceptual lesson remains relevant: different graph observables can respond differently to the same edge rule and null model, so a complete audit should eventually vary both the diagnostic and the construction.

6 Discussion

The paired audit changes, but does not erase, the original result. $C_{\log}$ displays organized radial behavior on the black-hole embedding graphs. The question is whether the difference from a flat control is stable when the graph representation changes. It is not. Both the magnitude and the sign of the gap depend on the full protocol and on the random realization.

The most important correction is that radius-style controls do not absorb the signal in the canonical experiment. Strict-radius and edge-matched-radius controls instead produce the largest mean absolute gaps. Their near equality is especially informative: exact edge-count matching does not remove the effect, and a small residual edge-count difference in the strict-radius protocol does not explain it. At the same time, the broad seed-to-seed spread prevents a universal protocol ranking.

The result supports a more precise interpretation of $C_{\log}$. It is a statistic of finite-shell shortest-path multiplicity organization. Shortest paths are jointly determined by the sampled surface and by the imposed graph representation. The appropriate object of interpretation is therefore

$$\text{diagnostic} + \text{construction} + \text{control}, \quad (13)$$

not the diagnostic alone.

This distinction matters beyond black-hole embeddings. Geometric-graph studies should report the sampling prescription, graph scale, edge rule, shell radius, boundary handling, binning convention, seed distribution, and null-model construction. At minimum, a geometric claim should be tested against a sampling-matched control, a construction-matched control, and an edge-density or edge-count-matched control.

6.1 Scope and limitations

The present experiment is a finite-graph benchmark, not a continuum curvature reconstruction. It uses one graph size, one shell radius, one bin count, and ten paired realizations. The confidence intervals are broad, and the analysis does not establish continuum convergence. The graphs are built from Euclidean embeddings of spatial slices; relationships to intrinsic two-dimensional curvature, four-dimensional spacetime invariants, or non-embeddable data require separate analysis.

The Bardeen and Hayward results in this manuscript are representative fixed-protocol visual benchmarks. The five-protocol ten-seed audit has not yet been completed for those families, so the paper does not claim cross-family construction invariance or a universal ordering of protocols. Multi-resolution runs, shell-radius sweeps, alternative binning, and canonical audits of the additional diagnostics remain useful extensions.

6.2 Reproducibility

The audited calculation contains exactly 50 raw rows, ten rows per protocol and five rows per seed. All primary correlations are numeric, every run contains twelve common radial bins, and no seed is omitted. The maximum radial-angular pairing mismatch is 1.33×10^{-15} . All graphs are connected and all valid- $C_{\log}$ fractions equal one. The ten-seed computation required 3636.84 seconds of seed computation, with individual seeds taking 312.4–444.9 seconds.

Code, scripts, raw data, quality-control tables, checkpoints, validation files, and manuscript figures are available in the project repository. A DOI-backed archival snapshot should accompany journal submission.

7 Conclusion

We studied the shortest-path anisotropy statistic $C_{\log}$ on geometric graphs sampled from embedded black-hole spatial slices. Representative Schwarzschild/Flamm, Bardeen, and Hayward graphs show similar radial organization under the original calibrated construction. A paired ten-seed Schwarzschild/Flamm audit then tests how the black-hole/control separation changes under five graph/control protocols.

The audit does not confirm the earlier claim that radius-style controls erase the gap. Strict-radius and edge-matched-radius constructions instead produce the largest mean absolute separations, whereas the original and pure-KNN protocols produce smaller averages. Signed gaps vary across realizations and reverse under some protocols. The separation is therefore neither graph-construction invariant nor realization independent.

The main conclusion is methodological. $C_{\log}$ is a reproducible finite-shell shortest-path statistic under fully specified conditions, but it is not a graph-independent curvature scalar. Apparent geometric signals in spatial graph diagnostics should be interpreted as properties of complete diagnostic–construction–control protocols. Extending the same canonical paired audit to Bardeen and Hayward is the direct next test of how broadly this conclusion applies across black-hole embedding families.

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